Paleomagnetic Investigation of Some Arkansas Alkalic Igneous Rocks

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Remanent magnetization (RM) properties have been determined at nineteen sites in two alkalic igneous complexes at Magnet Cove and Potash Sulphur Spring, Arkansas (34.5°N, 29.8°W). The Magnet Cove intrusion has been dated by K-Ar, Rb-Sr, and fission track methods. The average age from six age dates is 98 m.y. (early Late Cretaceous). All specimens were demagnetized in various alternating peak fields and yield 19 site-mean RM directions. The paleomagnetic pole calculated from the mean of these 19 site-means is at 65.1°N and 173.3°W with confidence oval semiaxes 8.8° and 12.5°, respectively. This pole agrees with other North American Cretaceous poles and hence further supports the conclusion that the geomagnetic field during Late Cretaceous is dipolar and stable. It also suggests that central Arkansas has been stable since early Late Cretaceous.

INTRODUCTION

In the spring of 1967, samples were collected from alkalic igneous complexes at Magnet Cove and Potash Sulphur Spring in the central part of Arkansas (Figure 1). The original purpose was to study the relation between the magnetic properties and the lattice substitution of certain elements, such as chrominum, titanium, vanadium, and manganese. The preliminary results indicated, however, that these rocks were also suitable for paleomagnetic purposes. This brief report presents the paleomagnetic results obtained from these alkalic igneous complexes.

GEOLOGIC SETTING AND AGE RELATIONSHIPS

The alkalic igneous complexes occurring in the Magnet Cove and Potash Sulphur Spring areas are composed of a series of ring dikes that were intruded into folded and faulted Paleozoic sedimentary rocks of the Ouachita geosyncline. Extensive metamorphic contact zones surround the alkalic igneous complexes [*Erickson and Blade,* 1963]. Rock types involved in these alkalic igneous complexes, their absolute ages, and sample site distribution are listed in Table 1. The most common rock types, in the order of increasing age, are ijolite, melteigite, syenite, and jacupirangite. This relative age relation among rock types is based on the local geology of the Magnet Cove area [*Erickson and Blade,* 1963]. However, field relationships in the Potash Sulphur Spring area indicate that syenite is younger than ijolite [Pollock, 1965].

Six absolute ages from the Magnet Cove area have been reported and are listed in Table 1. Among them, three K-Ar and Rb-Sr dates are from melteigite and ijolite in the Magnet Cove area which give an average age of 97 m.y. [*Zartman et al.,* 1967]. The other three age dates from carbonatite and syenite were obtained by C. W. Naeser (personal communication, 1967) by using the fission track method. These yield an average age of 99 m.y., which agrees very well with the average age obtained by K-Ar and Rb-Sr method. Hence, the average age obtained from these six dates from the Magnet Cove area is 98 m.y., which indicates that the rocks in the Magnet Cove area were intruded during early Late Cretaceous (Cenomanian age as used in Europe). No absolute date has been reported for the Potash Sulphur Spring intrusion. However, similarities in the geology of the two areas suggest that the intrusions at Magnet Cove and Potash Sulphur Spring constitute a single event [*Hollingsworth,* 1967]. Therefore their ages, in general, are considered the same.

It should be noted that *Zartman et al.* [1967] also reported four other K-Ar and Rb-Sr dates from two nepheline syenites from Granite Mountain near Little Rock, about 40 miles northeast of the Magnet Cove area. The average of these dates is 88 ± 3 m.y. *Zartman et al.* [1967] concluded, 'These ages from Little Rock

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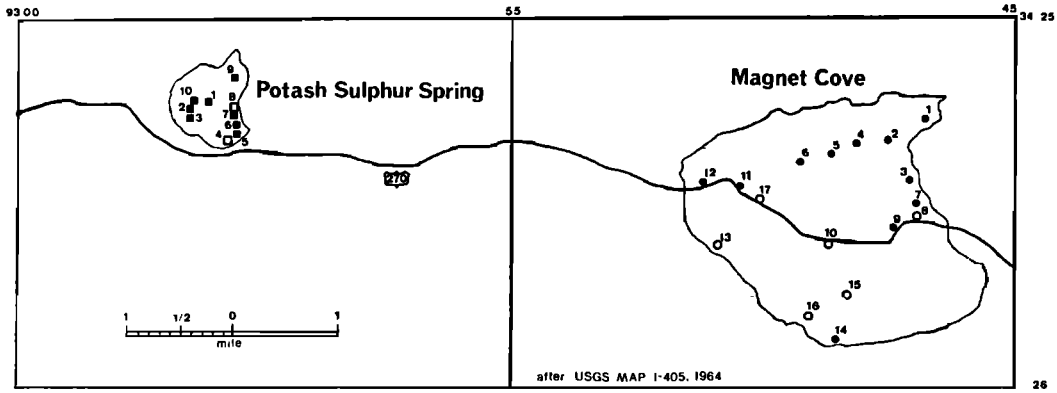


Fig. 1. Map showing paleomagnetic sampling sites of Arkansas alkalic igneous complex. Open symbols are discarded sites.

and Magnet Cove lie near the limits of analytical uncertainty of each other, and, therefore, they may represent contemporaneous or slightly different time of intrusion.' When comparing these four dates with the fission track dates, however, we note that the two intrusions may differ in time, since for syenite, the range of fission track dates from the Magnet Cove area does not overlap the range of the radiogenic dates from the Granite Mountain region. Therefore, the Granite Mountain syenite should be a later intrusion.

Paleomagnetic Results

Samples for studying magnetic properties were collected, by using a portable drill, from 17 sites in the Magnet Cove area and 10 sites in the Potash Sulphur Spring area (34.5°N, 29.8°W), and a total of 168 one-inch core specimens was prepared in the laboratory. For paleomagnetic purposes, four specimens from each site were used (with the exception of two sites M8 and P4). Natural remanent magnetization (NRM) of these specimens was measured by a spinner magnetometer. The averaged NRM results for

TABLE 1. Paleomagnetic Sample Sites and Their Related Rock Types and Age Dates

Paleomagnetic Sample Site	Rock Type	Age Dates, m.y.		
		K-Ar	Rb-Sr	Fission Track
P6	Tinguaite			
(M17)	Carbonatite			90 ± 9 (A)
				103 ± 10 (A)
P1	Calcar. Breccia			
P2, P3, (P4), P5	Syenite (P)			
M6 (M10), P7	Ijolite	97 ± 5 (BM)	99 ± 8 (BM)	
(P8), P9, P10	Melteigite	95 ± 5 (BM)		
M7, (M8), M11, M12, (M13), M14, (M16)	Syenite (M)			105 ± 10 (S)
M1, M2	Jacupirangite			
M4, M9, (M15)	Trachyte			
M3, M5	Altered Phonolite			

Notes.

(BM), Biotite from the Magnet Cove Area [*Zartman et al.* 1967]; (A), Apatite; (S) Sphene [*C. W. Naeser*, personal communication, 1967]; M, Magnet Cove area; P, Potash Sulphur Spring area. Discarded sample sites are in parentheses.

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each site, calculated by *Fisher*'s [1953] method, are listed in Table 2. It is obvious that a certain degree of scattering of NRM directions exists within each site (indicated by α_{95} angles) and/or between sites (comparing the mean declinations and inclinations). Stability of NRM was therefore tested by applying alternating-field demagnetization. For each site, a specimen was chosen for stepwise demagnetization. The peak field for each step was set successively at 25, 50, 100, 200, 400, 600, and 800 oe. Results from these 27 demagnetization spectra indicated that for most sites a field between 50 and 200 oe was sufficient to remove the unstable component of magnetization and that only three sites needed higher demagnetization fields. To obtain better demagnetization results, the peak field used for partial demagnetization for each individual site

TABLE 2. Summary of Mean Remanent Magnetization Data

	NRM						After AF Demagnetization					
Site	N	D	I	α_{95}	$J \times 10^4$	H	N	D	I	α_{95}	$J \times 10^4$	C
M01	4	313.3	74.4	10.7	074.27	100	3	332.7	50.4	11.8	021.76	570
M02	4	354.0	57.9	5.8	041.33	100	3	341.1	64.5	10.2	014.13	
M03	4	336.1	30.5	61.2	000.87	800	3	346.3	53.5	18.5	000.27	
M4	4	353.8	59.4	104.9	034.24	200	2	333.4	58.7	90.1	002.13	565
M05	4	356.8	53.9	12.3	059.16	100	4	321.4	57.9	5.1	028.25	
M6	4	336.5	48.5	39.0	027.41	50	2	15.0	62.8	82.6	009.19	
M7	4	306.7	−30.4	64.5	122.58	50	3	318.2	−0.3	13.9	012.54	
(M8)	2	153.8	56.9	180.0	003.30	...	...	...	...	...	...	
M9	4	6.3	53.7	47.1	016.12	400	2	274.2	32.5	92.1	001.53	
(M10)	4	153.1	40.4	17.2	120.46	50	(4)	137.2	38.8	16.3	041.51	
M11	4	36.9	44.8	39.8	060.88	100	2	351.1	32.0	99.9	003.12	575
M12	4	229.9	69.7	31.1	009.83	100	3	270.9	69.8	27.7	005.91	490
(M13)	4	50.9	46.8	42.3	002.14	100	(2)	327.1	57.5	180.0	000.58	363
M14	4	279.2	54.8	28.1	013.69	100	2	323.4	58.4	54.2	001.78	
(M15)	4	111.5	20.5	36.4	215.63	100	(4)	112.0	18.4	26.9	079.81	
(M16)	4	183.3	49.5	35.4	021.54	100	(3)	210.8	40.9	69.4	009.35	
(M17)	4	289.7	49.9	40.9	021.12	100	(4)	209.2	65.8	97.9	002.30	
P01	4	304.4	39.0	16.4	125.78	100	3	325.7	47.4	16.2	009.58	320
P02	4	141.7	−72.7	24.2	003.21	200	3	351.6	64.8	36.3	000.27	
P03	4	351.1	49.7	16.8	012.86	200	1	341.4	49.9	0.0	000.34	
(P04)	3	307.0	72.2	33.8	015.38	...	...	...	...	...	...	
P05	4	282.9	59.9	91.1	006.15	200	1	327.8	57.4	0.0	000.69	
P06	4	348.3	53.3	13.5	007.34	200	2	339.4	64.2	31.6	002.67	575
P07	4	330.9	4.5	112.5	005.92	200	3	326.1	55.2	33.4	001.16	
(P08)	4	47.4	−14.4	31.6	0/0.52	100	(2)	355.3	24.3	180.0	001.10	
P09	4	348.5	50.3	118.5	002.01	800	2	326.3	47.4	35.9	001.66	
P10	4	355.7	59.7	66.8	013.09	100	3	342.0	56.0	4.8	018.29	
All M & P sites (25)*								330.2	62.0	13.6	10.8	
M sites (11)†								326.2	53.1	16.0	9.2	
P sites (8)†								334.1	55.6	6.0	4.3	
M & P sites (19)†								329.6	54.3	8.9	7.1	
M & P specimens (47)†								329.0	54.2	6.4	8.7	

Notes.

N indicates number of specimens per site (discarded site and N in parentheses).

D and I are declination and inclination in degrees.

J is magnetic intensity (emu/cc).

H is demagnetization peak field in oersteds.

α_{95} is the semiangle of 95% confidence cone [*Fisher*, 1953].

C is the Curie point temperature in degrees centigrade.

* With the discarded sites.

† Excluding the discarded sites.

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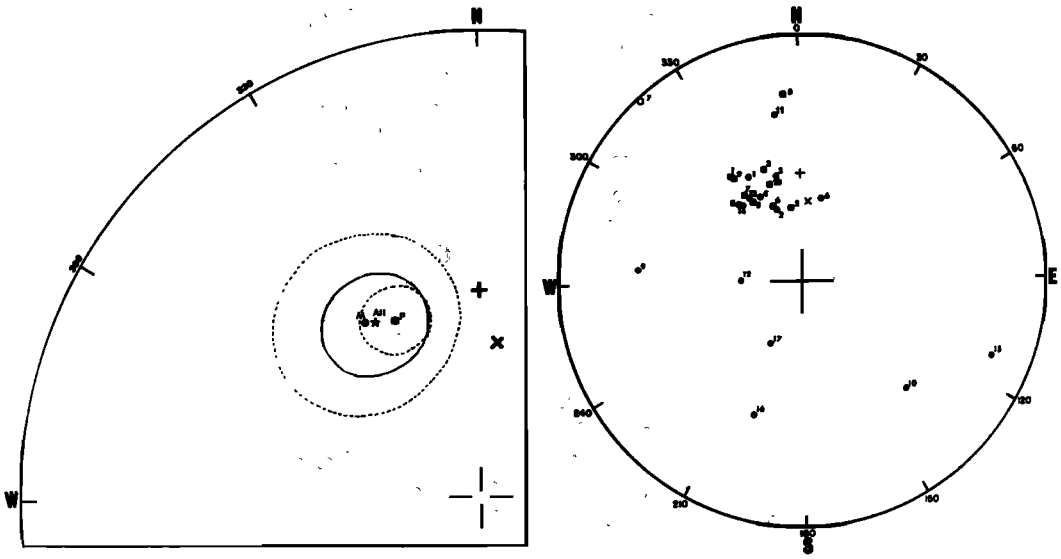


Fig. 2. Remanent magnetization (RM) direction of Arkansas alkalic igneous rocks: Site-mean directions after alternating field demagnetization (right) and group-mean directions as referred in Table 2 (left). Square, site in the Potash Sulphur Spring area; circle, site in the Magnet Cove area; solid symbol, north-seeking direction down; open symbol, south-seeking direction down; star, combined mean direction from 19 site-mean directions; oval. α_{95} confidence cone; horizontal cross, axial dipole-field direction; oblique cross, inclined dipole-field direction; Lambert equal area projection.

was determined by its demagnetization spectrum. The chosen demagnetization fields for each site are listed in Table 2.

After alternating-field demagnetization at various specified peak fields, only 66 specimens from 25 sites yielded results for further evaluation. The remaining 39 specimens were rejected because reliable remanent magnetization information could not be obtained from them. Criteria for rejecting these specimens were: (1) RM intensity after demagnetization was too weak to be measured (such as specimens from sites M13, P3, and P5), (2) the three measured RM components did not intersect to yield a reliable resultant vector, and (3) a specimen's RM direction deviated very much from the other specimens of the same site. By using data from these 66 after-demagnetization specimens, mean RM directions and intensities for each site were calculated. They are listed in Table 2. Among these 25 site-mean RM directions, it is obvious that sites M10, M15, M16, and M17 deviate very much from most of the other site-mean RM directions (Figure 2), and this is the reason that the over-all mean has a large α_{95} value (Table 2). Besides, the RM vectors of the two specimens in each of sites M13 and P8 are widely separated (see α_{95} angle in Table 2). These unreliable RM directions from the above six sites may be attributed to the secondary components which are hard enough to resist alternating-field demagnetization. With these six site-mean RM directions excluded, group-mean RM directions for the Magnet Cove area (11 sites), the Potash Sulphur Spring area (8 sites), and the two areas combined (19 sites) are listed in Table 2 and plotted in Figure 2. The insignificant difference in group-mean RM directions between the Magnet Cove intrusions and the Potash Sulphur Spring intrusions confirms that they are a single event, as the field relationship suggested. The combined mean RM direction from 19 site-means gives declination 329.6°, inclination 54.3°, and α_{95} 8.9°. It should be noted that only one specimen is available for each of sites P3 and P5. The question may be raised whether these two sites are overweighted in calculating the above combined mean direction. However, the mean direction given by 47 specimens (Table 2) is almost identical to the com-

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bined mean direction weighted by site-mean directions. It seems, therefore, that the method of weighting in the present case is not important.

Curie point measurements obtained by quartz-spring apparatus for sites M1, M4, P1, and P6 are 570°, 565°, 320°, and 575°C, respectively. For the specimens from site M12, three Curie point values were obtained by using high-frequency apparatus [*Petersen*, 1965] with a major peak at 575°C and minor peaks at 490° and 363°C. Curie point values ranging from 565° to 575°C for sites M1, M4, P6, and M12 indicate that magnetite is the chief magnetic mineral. Microscopic examination revealed that the low Curie point obtained from site P1 is explained by the presence of pyrrhotite as the main magnetic mineral, although no anomalous remanent magnetization was revealed. Lower Curie point peaks detected for specimens from site M12 may indicate the presence of minor elements in the structure of magnetite.

Pole Position

The virtual geomagnetic pole (VGP) calculated from the group-mean RM direction for 19 sites from both the Magnet Cove area and the Potash Sulphur Spring area is located in the vicinity of Bering Strait at 65.1°N, 173.3°W, with confidence oval semiaxes 8.8 and 12.5, respectively. This pole and seven other dated North American Late Cretaceous poles are plotted in Figure 3. Although these poles were obtained from widely separated (in the sense of both time and location) Late Cretaceous formations, the poles are not significantly different from each other, as indicated by their confidence ovals (Figure 3). Therefore, the present investigation further supports the following statements: (1) the hypothesis of a dipolar earth magnetic field during Late Cretaceous time is valid [*Grommé et al.*, 1967; *Larochelle*, 1968]; (2) the earth's magnetic field was stable during this period of time; i.e., no apparent polar wandering occurred [*Grommé et al.*, 1967]; and (3) the central part of Arkansas has been stable since Late Cretaceous time.

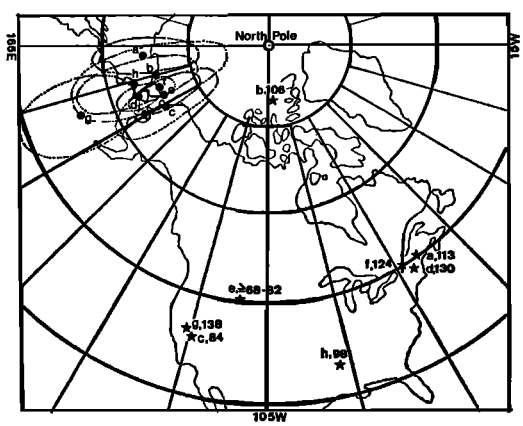


Fig. 3. Geographic distribution of eight North American Cretaceous poles (solid circles) and their corresponding sample sites (stars) with radiometric ages in millions of years. The 95% confidence ovals are shown as dotted lines. Poles are as follows: a, Mount Magnetic intrusive [*Larochelle*, 1961]; b, Isachsen diabase [*Larochelle et al.*, 1965]; c, Sierra Nevada granites [*Grommé and Merrill*, 1965]; d, Mount Ascutney intrusives [*Opdyke and Wensink*, 1966]; e, Elkhorn Mountain volcanics [*Hanna*, 1967]; f, Montergian Hills basic intrusives [*Larochelle*, 1968]; g, Guadalupe Mountain and Bucks plutons [*Grommé et al.*, 1967]; h, this paper.

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